Spatial Stats - Homework 6.3

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1. 7.5

Solution: For the power law model with generalized autocovariance $K_\phi(r) = \phi_1\Gamma(-\phi_2)r^{2\phi_2}$ on the 5×5 grid, with 24 linearly independent contrasts as differences between the center observation at $(0.5, 0.5)$ and each of the other 24 locations. Let D denote the 25×24 matrix whose columns are these contrast vectors, and let $C(\phi_2)$ be the 25×25 generalized covariance matrix with entries $C_{ij}(\phi_2) = K_\phi(\|s_i - s_j\|)$. The covariance matrix of the contrasts is then

$$\Sigma(\phi_2) = D^\top C(\phi_2)D,$$

which is positive definite of dimension 24×24.

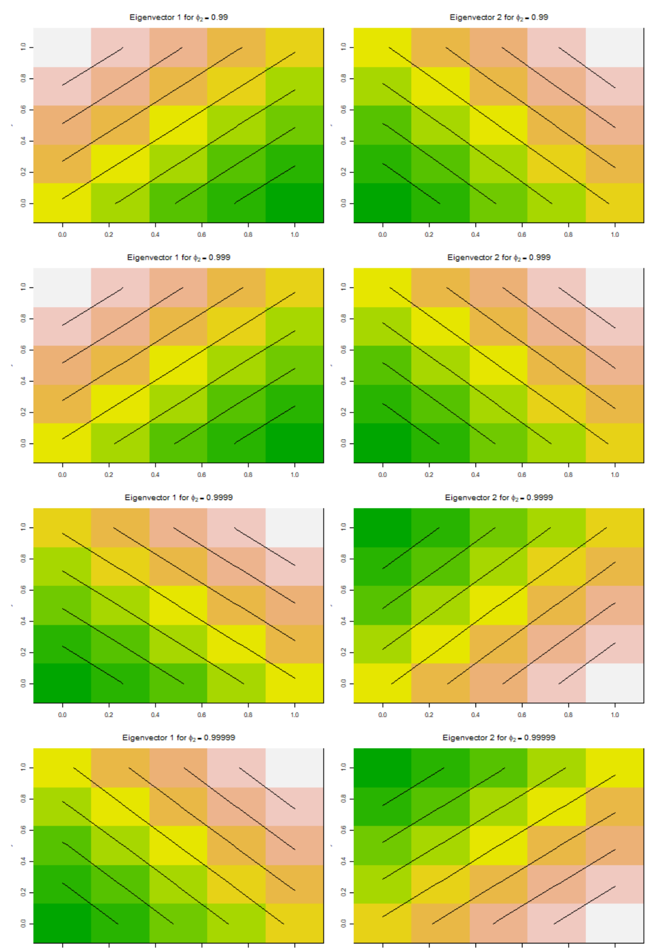
I computed the eigenvalues of $\Sigma(\phi_2)$ numerically for $\phi_2 = 1 - 0.1^k$ with $k = 2, 3, 4, 5$, i.e. for $\phi_2 \in \{0.99, 0.999, 0.9999, 0.99999\}$. The largest five eigenvalues are reported in Table 1.

ϕ_2	λ_1	λ_2	λ_3	λ_4	λ_5
0.99	625.296408	625.296408	12.171594	2.560734	1.355417
0.999	6250.232117	6250.232117	12.069184	2.556182	1.348082
0.9999	62500.225674	62500.225674	12.059151	2.555755	1.347365
0.99999	625000.200000	625000.200000	12.058150	2.555712	1.347294

Table 1: Largest eigenvalues $\lambda_1, \ldots, \lambda_5$ of the 24×24 contrast covariance matrix $\Sigma(\phi_2)$ for selected values of ϕ_2.

Solution: In all cases the largest two eigenvalues are equal, and they increase by approximately a factor of ten each time ϕ_2 moves closer to 1. Specifically, for $\phi_2 = 0.99$ I obtained $\lambda_1 = \lambda_2 \approx 6.25 \times 10^2$, while for $\phi_2 = 0.99999$ these grow to $\lambda_1 = \lambda_2 \approx 6.25 \times 10^5$, with the remaining 22 eigenvalues staying of order one. Thus, as $\phi_2 \to 1$ the eigenstructure becomes increasingly dominated by a two-dimensional leading eigenspace: almost all of the variance of the 24 contrasts is concentrated in two directions, while variation in the other 22 directions is comparatively negligible.

To interpret these dominant directions, I mapped the associated eigenvectors back onto the 5×5 grid (inserting a zero at the center location) and plotted them as images. For each ϕ_2 the first eigenvector shows a smooth gradient across the grid, increasing from one corner to the opposite corner, whereas the second eigenvector shows a similar gradient along the orthogonal diagonal. These patterns are essentially unchanged as ϕ_2 varies from 0.99 to 0.99999, indicating that the *shape* of the leading spatial modes is stable; only their eigenvalues change. Together with Table 1, this shows that near the boundary $\phi_2 \approx 1$ the power law model allocates almost all of the variance of the intrinsic contrasts to two large-scale gradient patterns over the grid, while higher-frequency or more localized contrast patterns (associated with the smaller eigenvalues) contribute relatively little. This behaviour reflects the strong long-range dependence of the power law covariance as ϕ_2 approaches 1.



Remark. Excerise 4

Solution: For a Gaussian process with power law generalized autocovariance and true value of ϕ_2 substantially less than 1, the profile restricted log-likelihood difference $pr\ell(0.99999) - pr\ell(0.9999)$ will tend to change very slowly as the number of observations n increases. A heuristic explanation can be given by examining how the covariance matrix depends on ϕ_2 near the boundary point $\phi_2 = 1$.

The generalized autocovariance has the form $K_\phi(r) = \phi_1 \Gamma(-\phi_2) r^{2\phi_2}$, so for any fixed distance $r > 0$ we have

$$K_\phi(r; \phi_2 + \delta) = K_\phi(r; \phi_2)\ \exp\bigl(2\delta \log r + o(\delta)\bigr)$$

as $\delta \to 0$. When ϕ_2 is already extremely close to 1, moving from $\phi_2 = 0.9999$ to $\phi_2 = 0.99999$ corresponds to a tiny increment $\delta = 9 \times 10^{-5}$, so each individual covariance entry changes only by a relative amount of order $\delta \log r$, which is very small for the fixed interpoint distances in a typical spatial sampling design. In matrix terms, the covariance matrices $C(0.9999)$ and $C(0.99999)$ differ by a small perturbation in operator norm.

For Gaussian data, the (restricted) log-likelihood can be written as

$$pr\ell(\phi_2) = -\tfrac{1}{2}\Bigl\{\log\det C(\phi_2) + Z^\top C(\phi_2)^{-1} Z\Bigr\} + \text{const},$$

after profiling out the mean and scale parameters. Since both $\log\det C(\phi_2)$ and $C(\phi_2)^{-1}$ are smooth functions of $C(\phi_2)$, a small perturbation of size δ in C produces a change in $pr\ell(\phi_2)$ of order δ^2 in a neighbourhood where the likelihood surface is reasonably flat. Thus the difference $pr\ell(0.99999) - pr\ell(0.9999)$ is itself of very small order in δ.

When the true value of ϕ_2 is substantially less than 1, both 0.9999 and 0.99999 lie far out in the tail of the likelihood, in a region where the data provide little information to distinguish between them. Increasing the sample size n refines the estimate of ϕ_2 in the interior of the parameter space but does not appreciably change the relative support for these two extreme boundary values, since they induce almost indistinguishable covariance structures. Consequently the vertical gap $pr\ell(0.99999) - pr\ell(0.9999)$ changes only very slowly as n grows.

2. 7.6

Solution: I compared the expected Fisher information for REML estimation under a two-parameter power law generalized covariance and a two-parameter exponential covariance, each with an unknown constant mean. For each model I fixed a parameter value and computed the 2×2 expected Fisher information matrix $I(\theta)$ for four different designs of $n = 25$ observation locations in the unit square:

- a regular 5×5 grid (`grid`);
- a slightly jittered version of the grid (`jitter`);
- 25 points sampled independently from $\mathrm{Unif}([0,1]^2)$ (`uniform`);
- a central cluster with points sampled from a bivariate normal centered at $(0.5, 0.5)$ and truncated to $[0,1]^2$ (`center`).

For the power law model I used $\phi_1 = 1$, $\phi_2 = 0.7$, and for the exponential model I used $\sigma^2 = 1$, $\rho = 0.3$.

For each design and model I formed the covariance matrix $\Sigma(\theta)$ and its partial derivatives with respect to the two covariance parameters. With design matrix $X = \mathbf{1}_n$, the expected REML Fisher information for parameters θ_a, θ_b is

$$I_{ab} = \tfrac{1}{2}\,\mathrm{tr}\{P\,\Sigma_{,a}\,P\,\Sigma_{,b}\}\,, \qquad P = \Sigma^{-1} - \Sigma^{-1}X(X^\top\Sigma^{-1}X)^{-1}X^\top\Sigma^{-1},$$

which I evaluated numerically for each design. To summarize the information matrices I recorded, for each 2×2 matrix $I(\theta)$, the trace $\mathrm{tr}(I)$, determinant $\det(I)$, and condition number $\kappa(I)$ (ratio of largest to smallest eigenvalue). The results are shown in Table 2 for the power law model and Table 3 for the exponential model.

Design	$\mathrm{tr}(I)$	$\det(I)$	$\kappa(I)$
grid	484.0	1.46×10^4	13.99
jitter	502.8	1.54×10^4	14.36
uniform	627.2	2.51×10^4	13.63
center	524.5	1.69×10^4	14.23

Table 2: Trace, determinant, and condition number of the Fisher information matrix for the power law covariance model under four sampling designs.

Design	$\mathrm{tr}(I)$	$\det(I)$	$\kappa(I)$
grid	80.1	3.25×10^2	17.69
jitter	80.7	3.34×10^2	17.48
uniform	98.3	3.28×10^2	27.43
center	130.8	1.79×10^2	93.31

Table 3: Trace, determinant, and condition number of the Fisher information matrix for the exponential covariance model under the same four sampling designs.

Solution:

For the power law model, both the trace and determinant of $I(\theta)$ vary substantially across designs. The trace increases from about 4.84×10^2 for the regular grid to 6.27×10^2 for the uniform design, while the determinant increases from about 1.46×10^4 to 2.51×10^4, indicating a marked gain in both marginal and joint precision of (ϕ_1, ϕ_2) when the design is changed.The central cluster design also differs appreciably from the grid, with intermediate trace and determinant values. In contrast, the condition numbers remain in a relatively narrow range (about 13.6–14.4), so the correlation structure between the two parameters does not change dramatically with design; the main effect is a sizeable change in the overall amount of information.

For the exponential model the picture is noticeably different. The trace increases from about 8.0×10^1 for the grid and jittered designs to 9.8×10^1 for the uniform design and 1.31×10^2 for the center design, a smaller relative change than in the power law case, the determinant stays close to 3.3×10^2 for the grid, jittered, and uniform designs, and only drops substantially for the highly clustered center design, indicating that the joint precision of (σ^2, ρ) is comparatively stable across most designs. While the condition number does become large for the center design, reflecting stronger correlation between the exponential parameters, the overall variation in $\mathrm{tr}(I)$ and $\det(I)$ across designs is smaller than for the power law model.

Taken together, these numerical experiments support the claim: the statistical properties of REML estimates (as reflected in the Fisher information) are much more sensitive to the choice of observation locations for the two-parameter power law covariance model than for the corresponding two-parameter exponential model.